Reinforcement Learning: In Theory and In Practice (ESE 3990)

Instructor:
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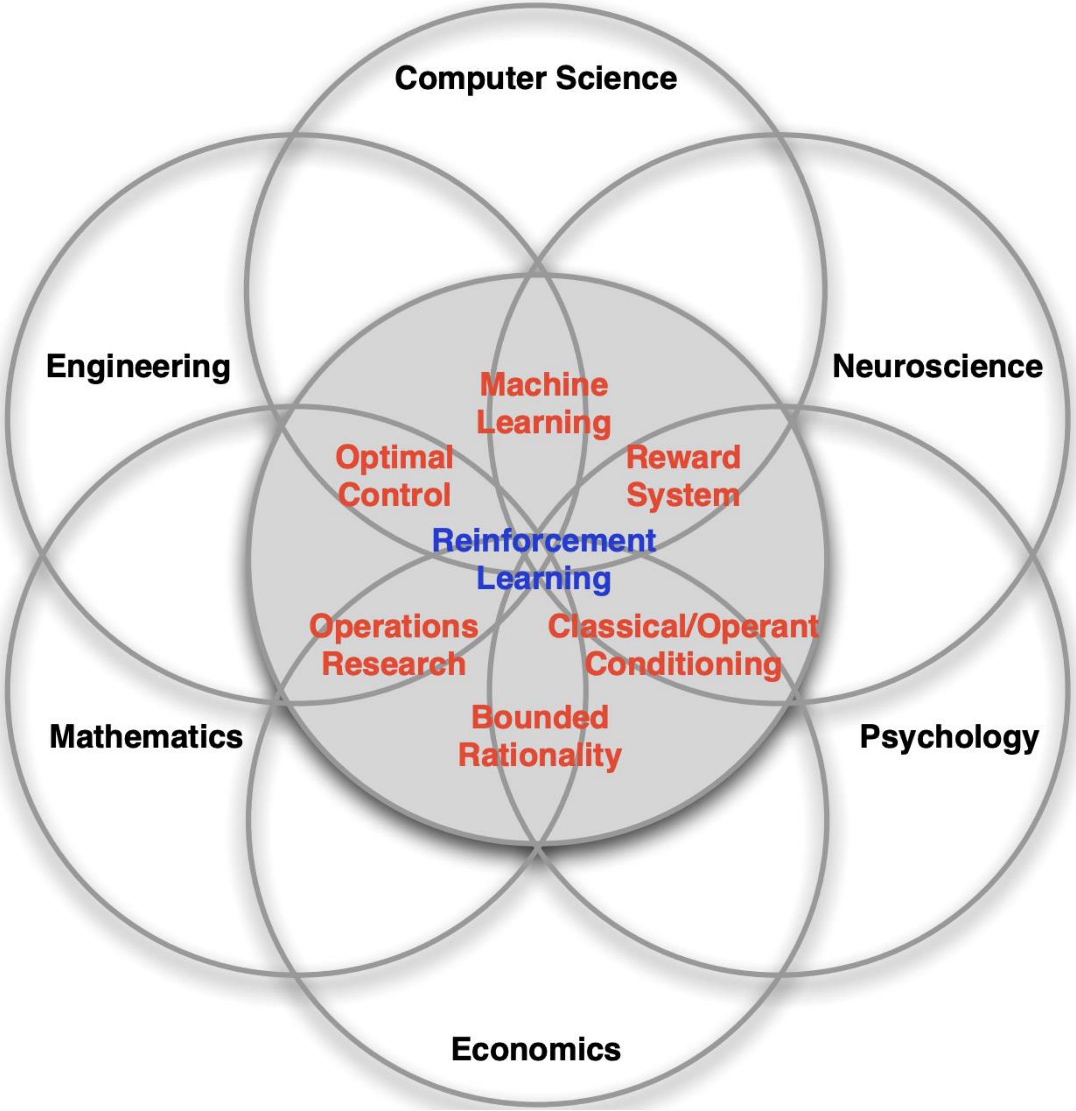
Fall 2026
University of Pennsylvania

Today

What is this class about?

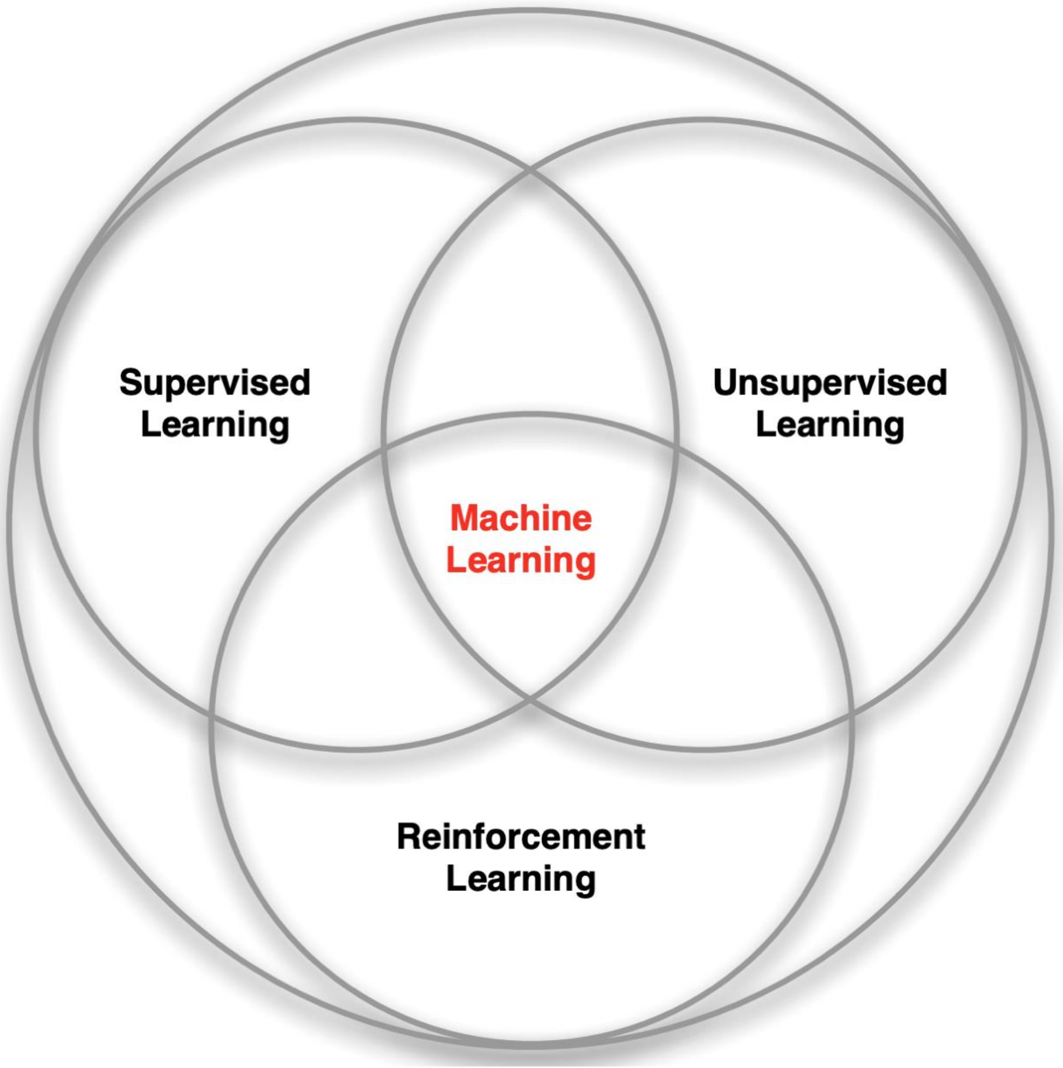
- What are you getting into?
- What are you getting out of it?
- What are the expectations?

Many Faces of Reinforcement Learning



Slide from D. Silver

Branches of Machine Learning



Slide from D. Silver

Characteristics of Reinforcement Learning

- What makes reinforcement learning different from other machine learning paradigms?
  - There is no supervisor, only a reward signal
  - Feedback is delayed, not instantaneous
  - Time really matters (sequential, non i.i.d data)
  - Agent’s actions affect the subsequent data it receives

Slide from D. Silver

Examples of Reinforcement Learning

- Outperform champion-level human pilots in drone racing, make a legged robot (quadruped or humanoid) run and walk.

- Defeat the world champion in Go

- Make the output of an LLM useful to users

- Control a power station

- Manage an investment portfolio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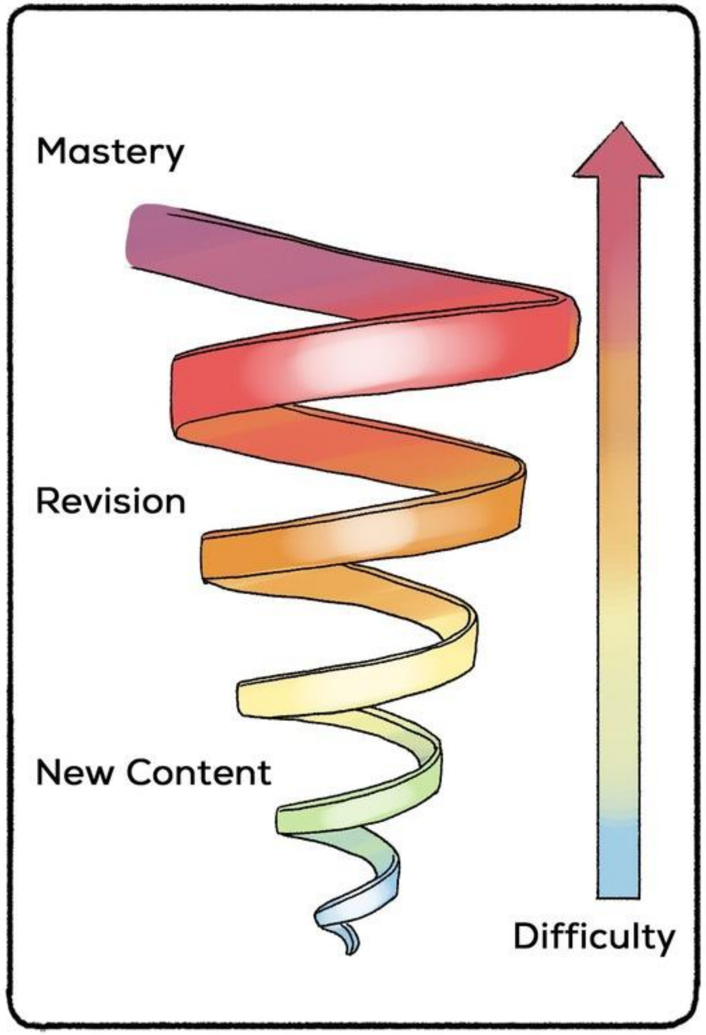
Talent
Awakening!

Course Structure

- Mathematical Foundations of RL
  - MDPs, Rewards, etc.
  - Value Functions
  - Policies
  - On-policy vs off-policy learning
- Reinforcement Learning Algorithms
  - Direct policy optimization (e.g., PPO)
  - Value-based methods (e.g., SAC and TD3)
  - Model-based methods (e.g., Dreamer and TD-MPC)
- Applications
  - Robotics
  - LLMs



The Spiral Teaching Method

What will you get out of this class

- Strong mathematical foundations of reinforcement learning
- Understand the differences between RL algorithms
- Nuanced understanding of the limitations of current methods and the frontier of research in the field

What will you **NOT** get out of this class

- Applications of RL/Supervised learning to Robotics
  - If you’re interested in this, please attend my ESE6510 class (Physical Intelligence).
- Implementation proficiency
  - The goal is that you understand RL algorithms, not how to implement them in the most efficient way possible.

Course Material & Tools

- All learning materials will be available on this website:
  - https://jirl-upenn.github.io/ESE3990_RL_Fall_26/

- We will use lot some tools to facilitate interactions:
  - Canvas for announcements
  - Ed for discussion
  - Gradescope for coursework

Course Organization

Grading:

1. Class Participation (10%)
    - **Attend class.** Average of top 3 quizzes**.** Ask questions, be involved!

2. Midterm (50%)

3. Project Presentation (40%)

Recitations

- We will review topics and do exercises together in class
- There will be a quiz during the recitation (approx. last 30min of each tutorial)
  - The participation score will be strongly related to the average top 3 scores you get in the quizzes.
  - Closed book (no laptop or phone)
- The midterm problems will be similar to the quizzes and problems in the recitation (just a tiny bit harder).
- No graded at-home exercises or homeworks.

Final Project

- You will get the opportunity to implement an RL algorithm *on your favorite problem*. Anything is good. Examples:
    - Control a simulated robot
    - Finetune a (small) LLM
    - Stock Market Optimization
    - …
- It is important that you code the algorithm yourself, **from scratch**.
    - The only way to actually learn the details.
    - Can use AI to help coding, but keep in mind the risks of doing so.
    - A comparison with a public/popular implementation would make for a great analysis (but not a requirement).

Final Project: Expectations

- You will do two presentations about the project.
  - In the first (~5min), you will present the idea and initial results. Use the feedback from the instructors and peers to improve!
  - In the second (~10min), you will present the final results and conclusions.

- We will provide you with a project presentation template.

- You will be graded according to:
  - Quality of presentation (including how you answer to questions!).
  - What you implemented (note: we will look at your code!).
  - The results you achieved.

Final Project: Example I

- Problem:
  - RL for Simulated Robot Control (Legged Robot)

- Project Goal:
  - Understand the hyperparameter sensitivity of off-policy algorithms.

- Experiments to run:
  - Verify your own implementation of the algorithm vs an existing library on a simple problem.
  - Integrate your algorithm in the simulated robot environment.
  - Run ablation studies on key hyperameters.
  - Form hypothesis on why some hyperparameters are more sensitive than others.

Final Project: Example II

- Problem:
  - RL for LLM Finetuning.
- Project Goal:
  - Finetune a text-based LLM with on-policy learning to maximize the value of a stock portfolio.
- Experiments to run:
  - Verify your own implementation of the algorithm vs an existing library on a simple problem.
  - Integrate your algorithm in the simulated stock environment.
  - Compare your algorithm to random stock selection.
  - Analyze what information the LLM is most sensitive to.

Final Project: Questions?

Opportunity to Increase Your Grade

- Take note, write them in latex, and share them with me.

- I’ll be writing notes for the full class at the end of this semester.

- If your notes make it to my notes, you will have the following perks:
  - Fame: Your name will be acknowledged in the chapter you contributed to.
  - Grade increase (e.g., from A to A+)
  - Gelato at the end of the class ☺

Some Polls!

- Poll of backgrounds
  - Machine learning experience?
  - Reinforcement learning experience?

If you haven't been admitted to the class yet, please register soon. Make sure to fill the form when you register (otherwise I can’t admit you).

Co-instructors

TA: Chunwei Xing

Guest Lecturer:
Avik De